

UMVUE analysis of two performance evaluation ratios

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ABSTRACT

We study the uniformly minimum-variance unbiased estimators (UMVUE) of two investment performance ratios in two different modelling settings: (1) the Sharpe ratio (SR) associated to the parameters of a geometric Brownian motion model, and (2) the information ratio (IR) associated to a factor model. In (1), in particular, we highlight how the estimation of the Sharpe ratio with log-returns from discrete sampling must take into account the variance (or Itô) correction in the log-price dynamics. In (2), we also derive the full distribution of the IR UMVUE, which is of interest in its own right. All methods in this note follow well-established foundational arguments of mathematical statistics and unbiased estimation; we will cite [1] often, as a graduate-level reference.

1. UMVUE of the continuous-time Sharpe ratio in GBM dynamics

Suppose that the price process $S = (S_t)_{t \geq 0}$ follows a geometric Brownian motion with strictly positive volatility $\sigma > 0$ and initial condition $S_0 = s_0 > 0$ and we want to estimate the Sharpe ratio (SR) of the continuous-time dynamics $dS_t/S_t = \mu dt + \sigma dW_t$ (i.e. the Sharpe ratio of the infinitesimal return):

$$\text{SR} := \frac{\mu}{\sigma}$$

The process S is sampled at equally-spaced observation times $\{0, \Delta t, 2\Delta t, \dots, k\Delta t, \dots\}$ where $\Delta t > 0$. A standard argument (indeed: apply Itô's formula to $f(S_t) = \ln S_t$) yields log-returns $X_k := \ln(S_{k\Delta t}/S_{(k-1)\Delta t})$, $k \in \mathbb{N}$ with $X_k \sim \mathcal{N}((\mu - \sigma^2/2)\Delta t, \sigma^2\Delta t)$, IID. The variance correction $-\sigma^2\Delta t/2$ makes the standard Sharpe ratio UMVUE (see e.g. ([1]; Ex. 3.4 p. 164)) imprecise, yielding only the unbiased estimator of $(\mu - \sigma^2/2)/\sigma$. For a random sample of $n \geq 4$ log-returns, define the sample mean $\bar{X}_n$ and the sample variance S_n^2 :

$$\begin{aligned} \bar{X}_n &:= \frac{1}{n} \sum_{1 \leq k \leq n} X_k \implies \bar{X}_n \sim \mathcal{N}\left((\mu - \sigma^2/2)\Delta t, \frac{\sigma^2\Delta t}{n}\right) \\ S_n^2 &:= \frac{1}{n-1} \sum_{k \leq n} (X_k - \bar{X}_n)^2 \implies \frac{(n-1)S_n^2}{\sigma^2\Delta t} \sim \chi_{n-1}^2 \end{aligned}$$

Then, $\bar{X}_n$ and S_n^2 are independent. Indeed, define $Y_k := X_k + \sigma^2 \Delta t / 2$. The sample mean $\bar{Y}_n$ is complete and sufficient for μ when σ^2 is known (because $\bar{Y}_n \sim \mathcal{N}(\mu \Delta t, \sigma^2 \Delta t / n)$, see e.g. ([1]; Ex. 2.18 p. 112)), while S_n^2 is ancillary for σ^2 fixed: then, $\bar{Y}_n$ is independent of S_n^2 by Basu's theorem, hence also $\bar{X}_n = \bar{Y}_n - \sigma^2 \Delta t / 2$. We can now compute the expectation of the natural candidate:

$$E \left[\frac{\bar{X}_n + S_n^2 / 2}{S_n} \right] = \Delta t \left(\mu - \frac{\sigma^2}{2} \right) E[S_n^{-1}] + \frac{1}{2} E[S_n]$$

We have:

$$\begin{aligned} E[S_n^{-1}] &= \frac{\sqrt{n-1}}{\sigma \sqrt{\Delta t}} E \left[\left(\frac{(n-1)S_n^2}{\sigma^2 \Delta t} \right)^{-1/2} \right] \\ &= \frac{\sqrt{n-1}}{\sigma \sqrt{\Delta t}} \int_0^\infty x^{-1/2} \frac{x^{(n-1)/2-1} e^{-x/2}}{2^{(n-1)/2} \Gamma((n-1)/2)} dx \\ &= \frac{\sqrt{n-1}}{\sigma \sqrt{\Delta t}} \int_0^\infty \frac{x^{(n-2)/2-1} e^{-x/2}}{2^{(n-1)/2} \Gamma((n-1)/2)} dx \\ &= \frac{\sqrt{n-1}}{\sigma \sqrt{\Delta t}} \frac{2^{-1/2} \Gamma((n-2)/2)}{\Gamma((n-1)/2)} \\ &= \frac{1}{\sigma \sqrt{\Delta t}} \frac{2^{-1/2} \Gamma((n-2)/2)}{(n-1)^{-1/2} \Gamma((n-1)/2)} \\ E[S_n] &= \sigma \sqrt{\Delta t} \frac{(n-1)^{-1/2} \Gamma(n/2)}{2^{-1/2} \Gamma((n-1)/2)} \end{aligned}$$

So that

$$E \left[\frac{\bar{X}_n / C_n + S_n^2 / (2D_n)}{S_n} \right] = \sqrt{\Delta t} \left(\mu - \frac{\sigma^2}{2} \right) \frac{1}{\sigma} + \frac{\sigma \sqrt{\Delta t}}{2} = \frac{\mu}{\sigma} \sqrt{\Delta t}$$

where

$$C_n := \frac{2^{-1/2} \Gamma((n-2)/2)}{(n-1)^{-1/2} \Gamma((n-1)/2)}, \quad D_n := \frac{(n-1)^{-1/2} \Gamma(n/2)}{2^{-1/2} \Gamma((n-1)/2)}$$

So we can conclude that the UMVUE of the Sharpe ratio μ/σ is

$$\boxed{\hat{\text{SR}}(X_1, \dots, X_n) := \frac{1}{S_n \sqrt{\Delta t}} \left(\frac{\bar{X}_n}{C_n} + \frac{S_n^2}{2D_n} \right), \quad n \geq 4}$$

since $(\bar{X}_n, S_n^2)$ is complete and sufficient, by Lehmann-Scheffé ([1]; Th. 3.1 p. 162). Note that $n \geq 4$ is needed for $V[\hat{\text{SR}}]$ to be finite.

2. UMVUE of the information ratio in a linear factor model

Consider the factor model specification $Y_k = \alpha + \beta X_k + \varepsilon_k$, $\varepsilon_k \sim \mathcal{N}(0, \sigma^2)$, IID where $X_1, X_2, \dots, X_n$ for $n \geq 5$ are known covariates and $\sigma > 0$. The information ratio (IR)

of the model is defined as

$$\text{IR} := \frac{\alpha}{\sigma}$$

Define the covariates sample variance $S_{X,n}^2$ (assumed strictly positive), the regression estimators $\hat{\alpha}_n, \hat{\beta}_n$ and the (scaled) residual sum-of-squares $\tilde{S}_n^2$:

$$\begin{aligned} S_{X,n}^2 &:= \frac{1}{n-1} \sum_{1 \leq k \leq n} (X_k - \bar{X}_n)^2 \\ \hat{\beta}_n &:= \frac{1}{(n-1)S_{X,n}^2} \sum_{1 \leq k \leq n} (X_k - \bar{X}_n)(Y_k - \bar{Y}_n) \\ \hat{\alpha}_n &:= \bar{Y}_n - \hat{\beta}_n \bar{X}_n \\ \tilde{S}_n^2 &:= \frac{1}{n-2} \sum_{1 \leq k \leq n} (Y_k - \hat{\beta}_n X_k - \hat{\alpha}_n)^2 \implies \frac{(n-2)\tilde{S}_n^2}{\sigma^2} \sim \chi_{n-2}^2 \end{aligned}$$

The regression estimators are unbiased estimators of α, β and it is well-known that $\hat{\alpha}_n$ (and $\hat{\beta}_n$ as well) is independent of $\tilde{S}_n^2$ (see e.g. ([1]; Th. 3.8 p. 188)). We get $\sqrt{n-2}\tilde{S}_n/\sigma \sim \chi_{n-2}$, hence we can evaluate the expectation of the natural candidate:

$$\begin{aligned} E \left[\frac{\hat{\alpha}_n}{\tilde{S}_n} \right] &= \alpha E[\tilde{S}_n^{-1}] \\ &= \frac{\alpha \sqrt{n-2}}{\sigma} \int_0^\infty x^{-1} \left(\frac{x^{(n-2)-1} e^{-x^2/2}}{2^{(n-2)/2-1} \Gamma((n-2)/2)} \right) dx \\ &= \frac{\alpha \sqrt{n-2}}{\sigma} \int_0^\infty \left(\frac{x^{(n-3)-1} e^{-x^2/2}}{2^{(n-2)/2-1} \Gamma((n-2)/2)} \right) dx \\ &= \frac{2^{(n-3)/2-1} \Gamma((n-3)/2)}{2^{(n-2)/2-1} \Gamma((n-2)/2)} \frac{\alpha \sqrt{n-2}}{\sigma} \\ &= \sqrt{\frac{n-2}{2}} \frac{\Gamma((n-3)/2)}{\Gamma((n-2)/2)} \frac{\alpha}{\sigma} \end{aligned}$$

So we conclude that the UMVUE of the information ratio α/σ is

$$\boxed{\hat{\text{IR}}(X_1, \dots, X_n; Y_1, \dots, Y_n) = \sqrt{\frac{2}{n-2}} \frac{\Gamma((n-2)/2)}{\Gamma((n-3)/2)} \frac{\hat{\alpha}_n}{\tilde{S}_n}, \quad n \geq 5}$$

again by Lehmann-Scheffé ([1]; Th. 3.1 p. 162). Note that an entirely equivalent argument holds for the UMVUE of β/σ . The restriction $n \geq 5$ is needed for $V[\hat{\text{IR}}]$ to be finite. We shall now derive the finite-sample distribution of $\hat{\text{IR}}$. Define

$$\begin{aligned} F_n &:= \sqrt{\frac{n-2}{2}} \frac{\Gamma((n-3)/2)}{\Gamma((n-2)/2)} \\ \gamma_n^2 &:= \frac{1}{F_n^2} \left(\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2} \right) \end{aligned}$$

We can then rewrite the UMVUE as

$$\hat{\text{IR}}(X_1, \dots, X_n; Y_1, \dots, Y_n) = \frac{\hat{\alpha}_n}{\sigma F_n} \Big/ \left(\frac{1}{\sqrt{n-2}} \frac{\tilde{S}_n \sqrt{n-2}}{\sigma} \right)$$

where

$$\frac{\hat{\alpha}_n}{\sigma F_n} \sim \mathcal{N}\left(\frac{\alpha}{\sigma F_n}, \gamma_n^2\right)$$

So we get the noncentral t -distribution

$$\frac{\hat{\text{IR}}(X_1, \dots, X_n; Y_1, \dots, Y_n)}{\gamma_n} = \frac{\hat{\alpha}_n / \tilde{S}_n}{\sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}} \sim t\left(n-2, \frac{\alpha}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}}\right)$$

Which ultimately implies

$$\hat{\text{IR}}(X_1, \dots, X_n; Y_1, \dots, Y_n) \sim \frac{\sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}}{F_n} \cdot t\left(n-2, \frac{\alpha}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}}\right)$$

that is, the UMVUE has a scaled noncentral t -distribution. Sanity check: for $n \geq 5$

$$E[\hat{\text{IR}}(X_1, \dots, X_n; Y_1, \dots, Y_n)] = \frac{\sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}}{F_n} F_n \frac{\alpha}{\sigma \sqrt{\frac{1}{n} + \frac{\bar{X}_n^2}{(n-1)S_{X,n}^2}}} = \frac{\alpha}{\sigma}$$

which is consistent with what we have shown above.

References

- [1] Shao, J. (2003). *Mathematical Statistics (Second Edition)*. Springer-Verlag New York.